Tutorial Questions | Week 10

COSC2779 - Deep Learning

This tutorial reviews deep generative models: autoencoders and variational autoencoders, generative adversarial networks, and diffusion models. Please try the questions before you join the session. The question below has a similar structure to the case study questions that you can expect in the end semester test.

1. You have been hired by a hospital research group that wants to build a **generative model of chest X-ray images**. They have a large set of unlabelled images. They have several goals in mind: (i) detecting unusual (anomalous) scans, (ii) **image retrieval** – given a query scan, find the most visually similar scans in the archive – and later (iii) generating realistic synthetic scans to augment a small labelled dataset for a downstream classifier. You will work through the main families of generative models to decide what fits.

   (a) You first train a plain **autoencoder** (encoder → bottleneck → decoder) on the images with a reconstruction loss. Explain how the same trained model can be used for goals (i) anomaly detection and (ii) image retrieval, and name a suitable reconstruction loss.

   (b) You now try to generate new scans by sampling a random point in the autoencoder's bottleneck and decoding it – but you get unrealistic junk. Why does a plain autoencoder fail as a generator?

   (c) A **variational autoencoder (VAE)** fixes this. What two changes does it make, and what is the role of each term in its loss $\mathcal{L} = \underbrace{\mathbb{E}[-\log p(x|z)]}_{\text{reconstruction}} + \underbrace{D_{KL}(q(z|x) \,\|\, \mathcal{N}(0, I))}_{\text{KL}}$?

   (d) Training a VAE needs to back-propagate through a sampling step $z \sim \mathcal{N}(\mu, \sigma^2)$. What trick makes this possible, and why is it needed?

   (e) VAE samples come out realistic but **blurry**. A colleague suggests a **GAN** instead. Describe the two networks and the objective they optimise.

   (f) During GAN training you notice the generator only ever produces a couple of very similar scans. What is this failure called, and give one reason it happens.

   (g) You move to a **diffusion model**. Describe its **forward** process and its **reverse** process, and say which one is learned.

   (h) In the DDPM formulation, what does the network actually predict, and what loss is used? Why is this attractive compared to a GAN?

   (i) What kind of network is typically used for ϵ_θ on images, and why is that architecture a good fit? What is the main practical drawback of diffusion at generation time?

   (j) For goal (iii) the group needs to generate scans of a specific finding (e.g. "pneumonia"). How would you make any of these generators produce a chosen class rather than a random sample?

   (k) Summarise the trade-offs: rank VAE, GAN, and diffusion on (a) training stability, (b) sample quality/diversity, and (c) sampling speed. Which would you recommend here, and why?